Introduction to Limits

ANSWER KEY

Evaluating limits

1 Evaluate $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression.
$\frac{(x-2)(x+2)}{x-2} = x + 2$ for $x \neq 2$, so the limit is 4.

2 Evaluate by direct substitution where possible:

a $\lim_{x \to 3} (x^2 + 2x - 1) \quad = 14$

b $\lim_{x \to 1} \frac{x+5}{x+1} \quad = 3$

3 Using a table of values near $x = 0$ (in radians), estimate $\lim_{x \to 0} \frac{\sin x}{x}$.
Values approach 1 from both sides; the limit is 1.

One-sided limits and limits at infinity

4 Let $f(x) = \frac{|x|}{x}$. Find:

a $\lim_{x \to 0^+} f(x) \quad = 1$

b $\lim_{x \to 0^-} f(x) \quad = -1$

c $\lim_{x \to 0} f(x)$ does not exist

5 Evaluate each limit at infinity:

a $\lim_{x \to \infty} \frac{3x^2 + 5}{x^2 - 2} \quad = 3$

b $\lim_{x \to \infty} \frac{2x+1}{x^2 + 4} \quad = 0$

6 The graph of $f(x) = \frac{1}{x-2} + 3$ has asymptotes $x = 2$ and $y = 3$. Express each asymptote as a limit statement.
$\lim_{x \to 2^+} f(x) = +\infty$ (and $\lim_{x \to 2^-} f(x) = -\infty$); $\lim_{x \to \pm\infty} f(x) = 3$.

More limit computations

7 Evaluate $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$ by rationalizing the numerator.
Multiply by $\frac{\sqrt{x}+2}{\sqrt{x}+2}$: $\frac{x-4}{(x-4)(\sqrt{x}+2)} = \frac{1}{\sqrt{x}+2} \to \frac{1}{4}$.

8 Evaluate $\lim_{x \to -1} \frac{x^3 + 1}{x + 1}$ by factoring (using the sum-of-cubes formula).
$x^3 + 1 = (x+1)(x^2 - x + 1)$, so the limit equals $\lim_{x \to -1} (x^2 - x + 1) = 1 + 1 + 1 = 3$.

9 Evaluate $\lim_{x \to 0} \frac{\tan x}{x}$, using $\lim_{x \to 0} \frac{\sin x}{x} = 1$ and $\tan x = \frac{\sin x}{\cos x}$.
$\frac{\tan x}{x} = \frac{\sin x}{x} \cdot \frac{1}{\cos x} \to 1 \cdot \frac{1}{1} = 1$.

Continuity and limits together

10 A function f is defined by $f(x) = x + 3$ for $x \neq 2$, and $f(2) = 10$. Find $\lim_{x \to 2} f(x)$ and state whether f is continuous at $x = 2$.

$\lim_{x \to 2} f(x) = 2 + 3 = 5$. Since $f(2) = 10 \neq 5$, f is not continuous at $x = 2$ — it has a removable discontinuity.

11 Estimate $\lim_{x \to \infty} \left(\sqrt{x^2 + 1} - x\right)$ by rationalizing.

Multiply by $\frac{\sqrt{x^2+1}+x}{\sqrt{x^2+1}+x}$: $\frac{1}{\sqrt{x^2+1}+x} \to 0$ as $x \to \infty$.

Limits from graphs and tables

12 A function g satisfies $g(1.9) = 5.8$, $g(1.99) = 5.98$, $g(1.999) = 5.998$, $g(2.1) = 6.2$, $g(2.01) = 6.02$, $g(2.001) = 6.002$. Based on this table, estimate $\lim_{x \to 2} g(x)$.

Both sides approach 6: $\lim_{x \to 2} g(x) = 6$.

13 Explain, using one-sided limits, why $\lim_{x \to 0} \frac{1}{x^2}$ exists as an infinite limit (both one-sided limits agree), while $\lim_{x \to 0} \frac{1}{x}$ does not exist even as an infinite limit.

For $\frac{1}{x^2}$: as $x \to 0^+$ or $x \to 0^-$, $x^2 > 0$ always, so both one-sided limits are $+\infty$ — they agree, so we say $\lim_{x \to 0} \frac{1}{x^2} = +\infty$. For $\frac{1}{x}$: as $x \to 0^+$, $\frac{1}{x} \to +\infty$, but as $x \to 0^-$, $\frac{1}{x} \to -\infty$ — the one-sided limits disagree (not even both infinite in the same direction), so the limit does not exist in any sense.

Visualizing a limit graphically

14 The graph of $f(x) = \frac{x^2 - 4}{x - 2}$ (which simplifies to $x + 2$ with a hole at $x = 2$) is shown. Use the graph to estimate $\lim_{x \to 2} f(x)$, and explain why the hole doesn't affect the limit's value.

The graph approaches the height 4 from both sides as $x \to 2$, even though the function is undefined exactly at $x = 2$ (the hole). A limit only depends on the function's behavior NEAR the point, not its value (or lack of one) AT the point, so $\lim_{x \to 2} f(x) = 4$ regardless of the hole.

The formal idea of a limit (intuitive)

15 Explain, in your own words, what it means for $\lim_{x \to a} f(x) = L$, without using formal epsilon-delta language.

It means that as x gets arbitrarily close to a (from either side, without necessarily equaling a), the function's output values $f(x)$ get arbitrarily close to the single number L.

Limit laws

16 Given $\lim_{x \to 3} f(x) = 5$ and $\lim_{x \to 3} g(x) = -2$, use the limit laws to evaluate:

a $\lim_{x \to 3} [f(x) + g(x)] = 5 + (-2) = 3$

b $\lim_{x \to 3} [f(x) \cdot g(x)] = 5(-2) = -10$

c $\lim_{x \to 3} \frac{f(x)}{g(x)} = \frac{5}{-2} = -\frac{5}{2}$

Limits involving piecewise functions

17 A function is defined by $f(x) = x^2$ for $x < 1$ and $f(x) = 3x - 1$ for $x \geq 1$. Find $\lim_{x \to 1^-} f(x)$, $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.

$\lim_{x \to 1^-} f(x) = 1^2 = 1$. $\lim_{x \to 1^+} f(x) = 3(1) - 1 = 2$. Since the one-sided limits disagree ($1 \neq 2$), $\lim_{x \to 1} f(x)$ does NOT exist.